

Digital Coding Metasurfaces

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I. INTRODUCTION

Metamaterials are special structures engineered to provide properties that are not available in naturally available materials. Metasurfaces are the 2D versions of metamaterials. Metasurfaces are made up of multiple subwavelength structures (units) that have the ability to manipulate the amplitude, phase, polarisation of the light reflected off of them.

Digital coding metasurfaces were designed to control an EM waves by designing two distinct elements which have reflection phases separated by 180° at a given frequency. These two elements can now be mapped to the digital bits 0 and 1. A metasurface can be fabricated to have various combinations of these 0 and 1 unit cells. The radiation pattern of the reflected field can be controlled by controlling the coding pattern of the metasurface. This concept can be extended to a n-bit mapping where the reflection phases of consecutive bits are separated by $360/2^n$.

Usually, multiple of these elements in the same vicinity are given the same code to form a larger sub-region of dimensions $D_x \times D_y$. This is done so that the coupling between adjacent cells is negligible. We shall now call each of these sub-regions as one unit cell.

The state of each unit cell can be externally controlled using a FPGA (for example converting bit 0 to bit 1). The way in which this is done is by controlling the varactor diode. Our current paper does not delve into the implementation of this rather we just talk about proof of concept.

By externally controlling the coding pattern of the metasurface with time we can do what is called Space-Time Coding. Space-Time Coding leads to interesting properties which we will elaborate in later sections.

II. FAR FIELD PATTERN

A. Pattern Function

Consider the following metasurface consisting of $M \times N$ unit cells, each of dimensions $D_x \times D_y$ being illuminated by a normally incident plane wave. The electric field in the far-field region can be written as the vector sum of the individual contributions of each of the unit cells.

The electric field due the $(m, n)^{th}$ unit cell at a distance $r_{m,n}$ can be written as follows

$$E_{m,n} = E_{inc} \Gamma_{m,n} \frac{\exp(-jk_0 r_{m,n})}{r_{m,n}} \quad (1)$$

where E_{inc} is the magnitude of the incident electric field. $\Gamma_{m,n} = a_{m,n} \exp(j\phi_{m,n})$ is the reflection coefficient which

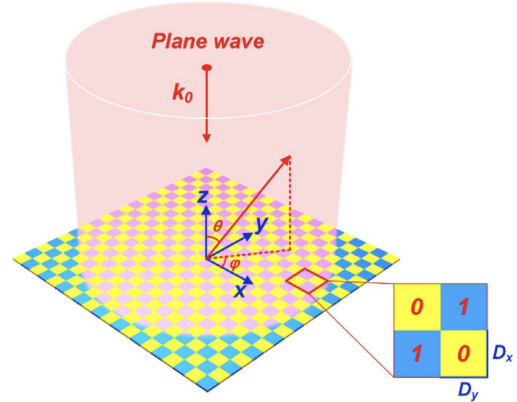


Fig. 1: A checkerboard coded metasurface illuminated normally by a plane wave (source:[1])

is composed of the multiplication of the amplitude and the phase reflection coefficients. For the sake of simplicity we shall assume that the amplitude reflection coefficient is always 1 and thus $\Gamma_{m,n} = \exp(j\phi_{m,n})$. A code of 0 corresponds to $\phi_{m,n} = 0$ and a code of 1 corresponds to $\phi_{m,n} = \pi$

Thus the net electric field in the far-field region at a distance of R from the metasurface can be written as

$$E(\theta, \varphi) = E_{inc} \frac{e^{-jk_0 R}}{R} \sum_{m=1}^M \sum_{n=1}^N \exp \left\{ j \left[\phi_{m,n} + k_o \times (\text{path length diff})_{m,n} \right] \right\} \quad (2)$$

that is, the pattern function (analogous to the array factor in antenna arrays) can be written as

$$F(\theta, \varphi) = \sum_{m=1}^M \sum_{n=1}^N \exp \left\{ j \left[\phi_{m,n} + k_o \times (\text{path length diff})_{m,n} \right] \right\} \quad (3)$$

The path length difference can be taking the dot product of the position vector of the unit cell

$$\mathbf{d}_{m,n} = \left(m - \frac{1}{2} \right) D_x \hat{\mathbf{x}} + \left(n - \frac{1}{2} \right) D_y \hat{\mathbf{y}} \quad (4)$$

with the unit vector along the direction of the point of interest in the far field.

$$\hat{\mathbf{u}} = \sin \theta \cos \varphi \hat{\mathbf{x}} + \sin \theta \sin \varphi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}} \quad (5)$$

This gives us the following final expression for the array factor-

$$F(\theta, \varphi) = \sum_{m=1}^M \sum_{n=1}^N \exp \left\{ j \left[\phi_{m,n} + k_0 D_x \left(m - \frac{1}{2} \right) \sin \theta \cos \varphi + k_0 D_y \left(n - \frac{1}{2} \right) \sin \theta \sin \varphi \right] \right\} \quad (6)$$

Using this equation for pattern function, we can plot the 3D polar plots for the far-field radiation pattern for any given code. Two examples have been shown below.

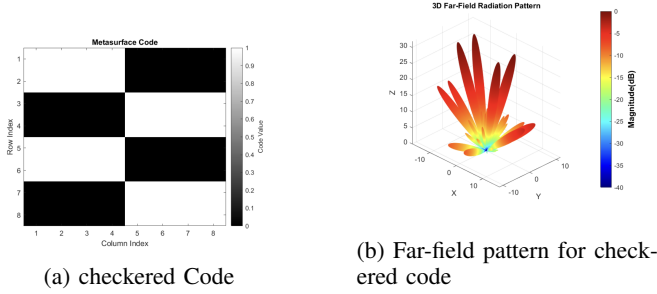


Fig. 2: Radiation pattern for checkered code

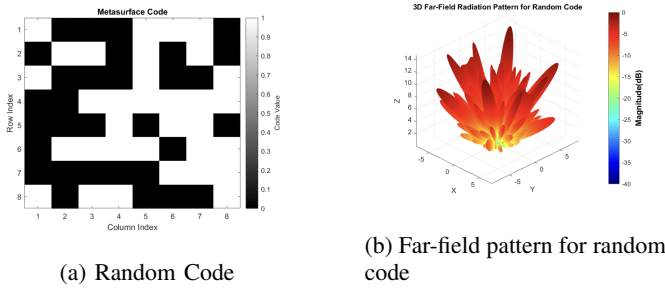


Fig. 3: Radiation pattern for random code

B. Code transform - analogy between pattern function and 2D DFT

The equation for the pattern function can be compared to the 2D DFT equation

$$X(k_x, k_y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x[m, n] e^{j(k_x m + k_y n)} \quad (7)$$

to get the following mapping between the pattern function equation and the 2D DFT equation -

$$\begin{aligned} F(\theta, \varphi) &\Longleftrightarrow X(k_x, k_y) \\ e^{j\phi_{m,n}} &\Longleftrightarrow x[m, n] \\ \phi_{m,n} &\Longleftrightarrow \angle x[m, n] \\ k_0 D_x \sin \theta \cos \varphi &\Longleftrightarrow k_x \\ k_0 D_y \sin \theta \sin \varphi &\Longleftrightarrow k_y \end{aligned} \quad (8)$$

On the basis of this analogy we introduce a new transform notation called the **Code Transform** to denote the pattern function.

$$F(\theta, \varphi) = CT(\Gamma_{m,n}) \quad (9)$$

The DFT is a transform between time domain and frequency domain and this new CT is a transform between code domain and far-field radiation pattern domain. We can use this code transform to explain the far-field radiation pattern of some special codes.

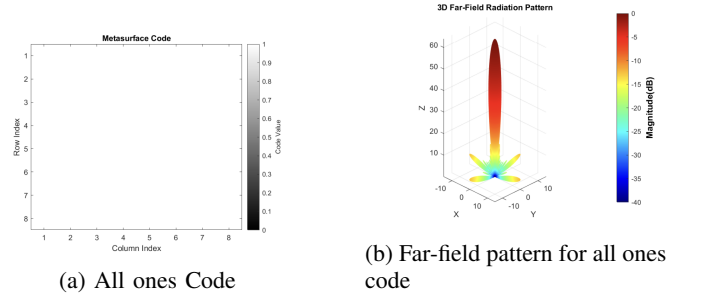


Fig. 4: Radiation pattern for all ones code

If we look at the far-field radiation pattern of a code of all ones, it looks like a very narrow beam. This reminds us of the well known fourier transform - $\mathcal{F}\{1\} = \delta(\omega)$.

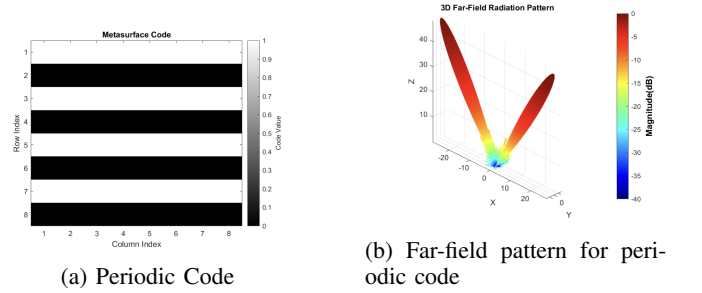


Fig. 5: Radiation pattern for periodic code

Similarly if we look at the radiation pattern of a code that is periodically varying only along one axis, there appear two narrow beam formations. This reminds us of another well known fourier transform - $\mathcal{F}\{\sin(\omega_0 t)\} = \frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$

C. Beam steering using convolution property

A multiplication in the time domain yields a convolution in the frequency domain. A convolution with a dirac delta at w_o implies a shift of the frequency spectrum to be centered at w_o , viz.,

$$f(t)e^{j\omega_0 t} \Longleftrightarrow F(\omega) * \delta(\omega - \omega_0) = F(\omega - \omega_0), \quad (10)$$

Now applying this same arguement to the code transform, we want to perform multiplication in the code domain to get a shift in the far-field domain, i.e a steering of the beam. A multiplication of exponentials is the addition of it's exponents,

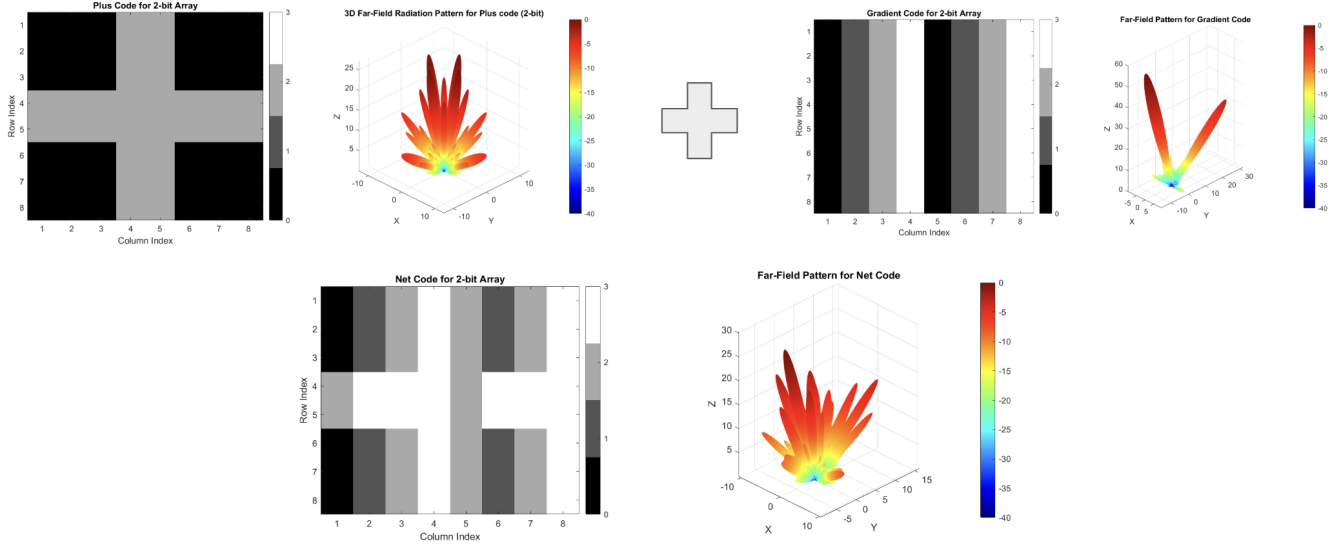


Fig. 6: Beam steering by exploiting the convolution property

so effectively we can add the phases of the code corresponding to the beam we want to steer to the code corresponding to a analogous dirac delta beam to result in a code which shall give a steered beam. We shall use a periodic gradient code to generate this narrow "dirac delta" beam.

We have performed this beam steering using 2 bit code as shown in Fig 6. The plus shaped code's phases are superimposed with gradient code. The resulting far-field pattern is a steered version of the plus code's far-field pattern.

III. SPACE TIME CODING

Now we shall analyse the effect of varying the metasurface code periodically with time on the far-field radiation pattern. The array factor for the time varying code can be written as:

$$F(\theta, \varphi) = \sum_{m=1}^M \sum_{n=1}^N \Gamma_{mn}(t) \exp \left\{ j \left[k_0 D_x \left(m - \frac{1}{2} \right) \sin \theta \cos \varphi + k_0 D_y \left(n - \frac{1}{2} \right) \sin \theta \sin \varphi \right] \right\} \quad (11)$$

where $\Gamma_{mn}(t)$ is the periodic time varying code for the (m,n) subregion. Since it is periodic we can write it as:

$$\Gamma_{mn}(t) = \sum_{l=1}^L \Gamma_{mn}^l U_{mn}^l(t) \quad (0 < t < T_0) \quad (12)$$

where $U_{mn}^l(t)$ is a periodic pulse function with modulation period T_0 . In each period, we have:

$$U_{mn}^l(t) = \begin{cases} 1, & (l-1)\tau \leq t \leq l\tau \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

where $\tau = T_0/L$ is the pulse width of $U_{mn}^l(t)$, L is a positive integer representing the length of the time-coding

sequence, and $\Gamma_{mn}^l = A_{mn}^l \exp(j\phi_{mn}^l)$ is the reflection coefficient of the (m,n)th coding element during the interval $(l-1)\tau \leq t \leq l\tau$ at the central frequency, with A_{mn}^l and ϕ_{mn}^l denoting the amplitude and phase, respectively.

Next, we decompose $U_{mn}^l(t)$ into a Fourier series:

$$U_{mn}^l(t) = \sum_{k=-\infty}^{\infty} c_{mn}^{lk} \exp(j2\pi k f_0 t) \quad (14)$$

where $f_0 = 1/T_0$, and the Fourier coefficients c_{mn}^{lk} are given by:

$$c_{mn}^{lk} = \frac{1}{T_0} \int_0^{T_0} U_{mn}^l(t) \exp(-j2\pi k f_0 t) dt \quad (15)$$

Thus, the Fourier series coefficients a_{mn}^k of the periodic function $\Gamma_{mn}(t)$ can be represented as:

$$\begin{aligned} a_{mn}^k &= \sum_{l=1}^L \Gamma_{mn}^l c_{mn}^{lk} = \sum_{l=1}^L \frac{\Gamma_{mn}^l}{T_0} \int_{(l-1)\tau}^{l\tau} e^{-j2\pi k f_0 t} dt \\ &= \sum_{l=1}^L \Gamma_{mn}^l \text{sinc} \left(\frac{k}{L} \right) \exp \left[\frac{-j\pi k (2l-1)}{L} \right] \end{aligned} \quad (16)$$

Finally, the far-field scattering pattern of the space-time-coding digital metasurface at the k th harmonic frequency $f_c + kf_0$, where k is any integer, is written as:

$$F_k(\theta, \varphi) = \sum_{m=1}^M \sum_{n=1}^N a_{mn}^k \exp \left\{ j \left[k_0 D_x \left(m - \frac{1}{2} \right) \sin \theta \cos \varphi + k_0 D_y \left(n - \frac{1}{2} \right) \sin \theta \sin \varphi \right] \right\} \quad (17)$$

Thus we conclude that the reflected field not only contains the incident wave frequency but also harmonic frequencies of $f_c + kf_0$, where k is any integer. We can also say that

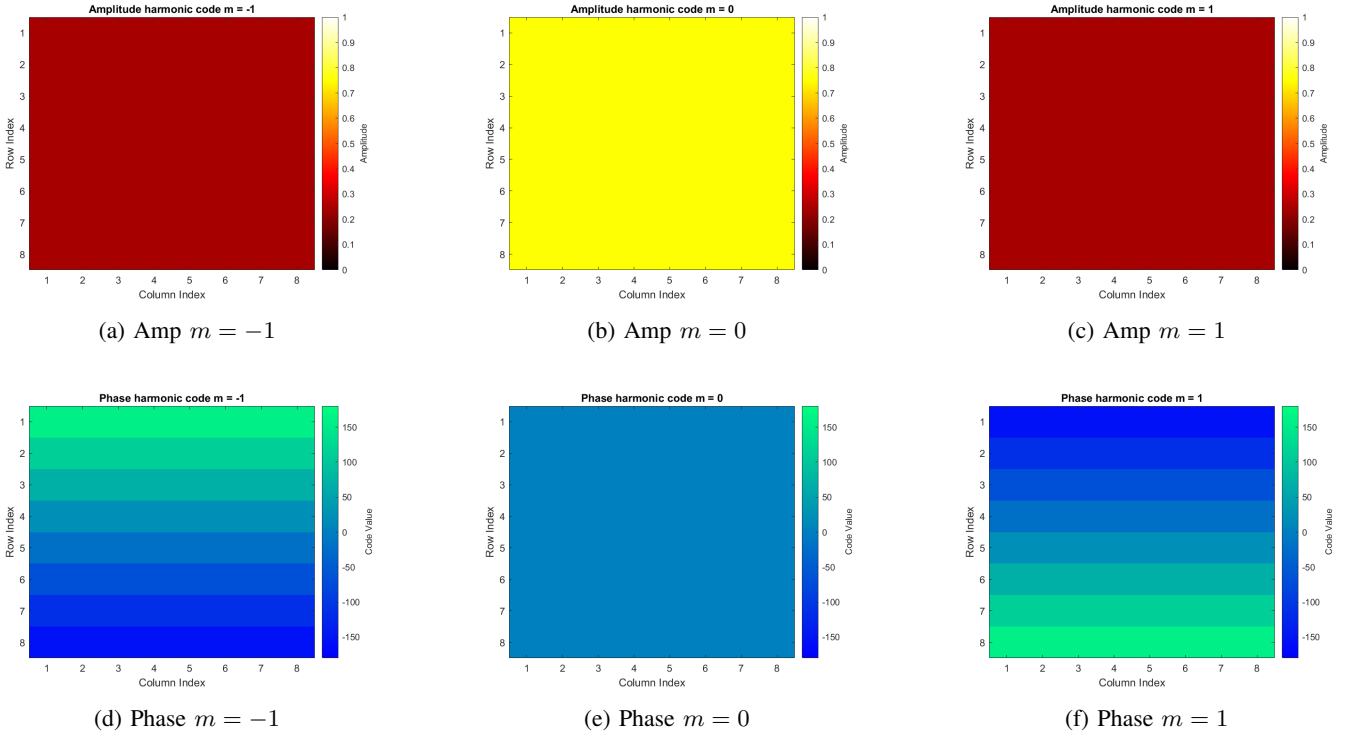


Fig. 7: Amplitude and Phase plots for the codes of varying values of m (Harmonic number). The $m = 0$ cases are shown in the center.

each frequency has an effective amplitude and phase response on the metasurface. This implies we can map each of these harmonics to their own effective static metasurface code by equating $\exp(j\phi_{m,n})$ in equation (6) to a_{mn}^k in equation (16).

A. Example Space Time Code

Let us look at an example of a space time code present in Fig. 8.

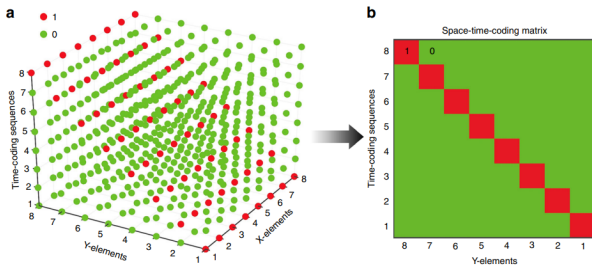


Fig. 8: Example space time code where each time instant a new row gets turned on. (source: [3])

We use the phases from this code to find the corresponding effective $a_{m,n}^k$ by plugging into equation (16). When we input this code into the formula we derived the results obtained are shown in Fig. 7. This clearly shows that with a particular space time code we can control the far field for each harmonic. This gives us immense control and has wide applications. We will now discuss two such applications.

IV. RCS REDUCTION

Our first application consists of a setting where we have an object (take it to be an aircraft) which we want to shield from a radar beam. In order to do this we can make some parts of the aircraft surface of metasurface material so that we can redirect the reflected wave in different directions. By doing so we can prevent the radar transmitter from getting back the reflected field. In order to motivate why space time coding does an extremely good job at this let us look at three different codes.

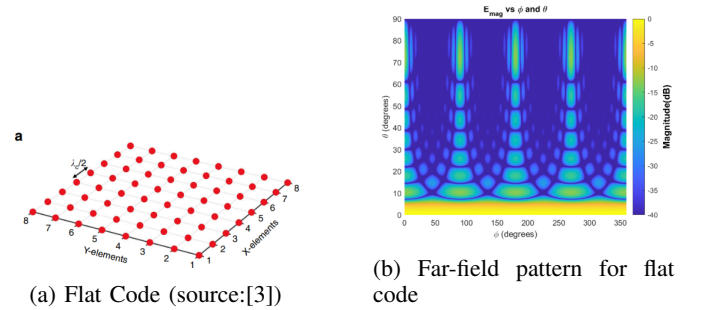
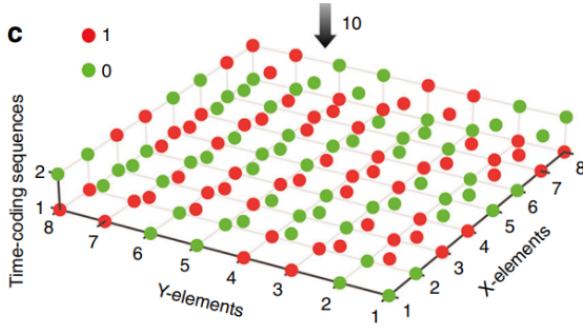


Fig. 9: Far field for flat code

The code in Fig.9 represents a normal metal surface with reflection coefficient -1 without any metasurface present. (Actually the reflection coefficient of metal is 1 but the reflected wave characteristics will be same.)



(a) Space-time coding configuration (source:[3])

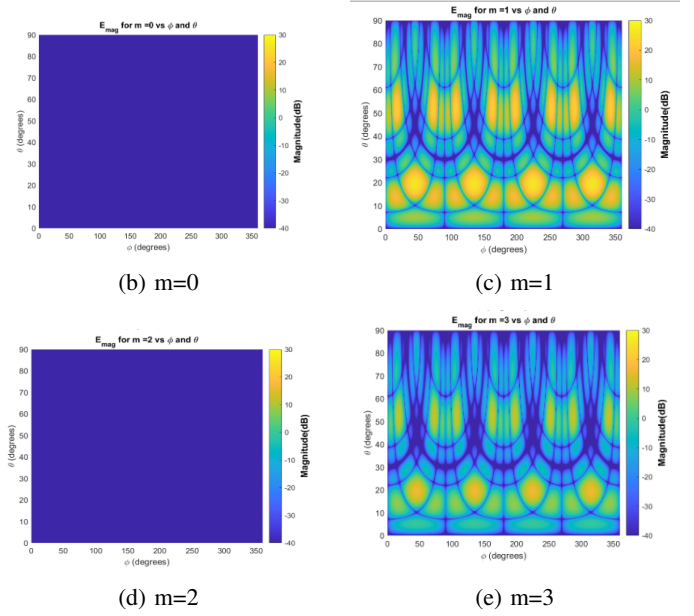


Fig. 10: Space-time modulation coding and resulting far-field harmonic distributions.

We can see in Fig.9 (b) the far-field for all θ, ϕ . The reflected far-field is highly concentrated at $\theta = 0$. In this configuration the radar antenna will receive back all its incident power.

V. FINE BEAM CONTROL

Another interesting application of space time coding is to get finer beam control. Let's say, for example, we have a 2-bit metasurface; we can only obtain reflection phases of $0, \pi/2, \pi, 3\pi/2$.

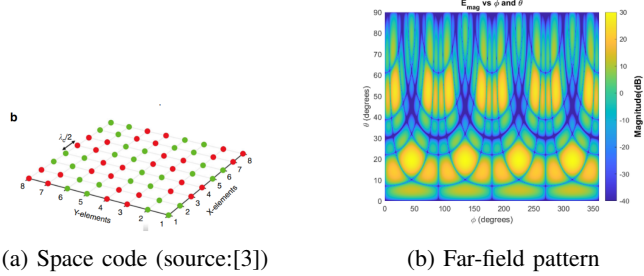


Fig. 11: Far field for space code

The code in Fig.11 is now a metasurface where we have done space coding. If we look at the far field generated by this you can see that the power is reflected in various direction and not right back to the transmitter. This is better than just a metal surface however it is not optimal as still many reflections can be retrieved by the radar transmitter.

Now let us look at a specific space time code in Fig.10. The advantage of space time coding is that you can not only reflect the incident wave spatially (in different directions) but also reflect the power onto different harmonics (frequencies). This dual degree of freedom of both spatial and frequency domains gives us an immense advantage. In the far field graphs you can see that for $m=0$ (which is the incident frequency) the reflected field is almost 0. The power is distributed to the odd harmonics.

By varying the space time code you can get the desired far field response for each harmonic.

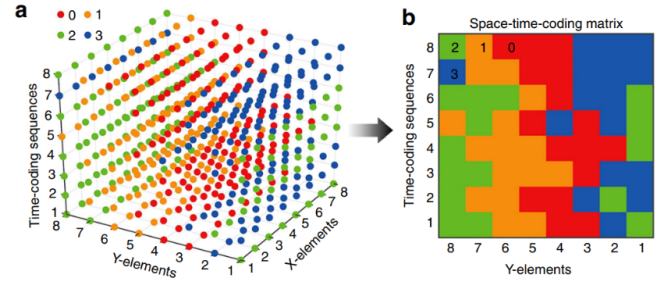


Fig. 12: Space Time code for fine beam control (source:[3])

Due to this we can only beam form to some particular precision. For example, let's consider the gradient code as shown in Fig.6. Different phase gradients shift the far field by different angles. Since the gradient code is restricted by the bit precision of the metasurface, we can only beam form to some particular angles. Space Time coding helps mitigate this issue by giving us access to more range of angles to steer the beam to.

Take Fig.12 as a particular space time code. If we use this code and plot the phase response for $m=0$ we get Fig.13. We see that it is a gradient but has phases other than what an equivalent 2-bit space code would have. We have now obtained better precision on beam forming just by varying our code with time. So rather than increasing the number of bits for



Fig. 13: Phase code for $m=0$ showing multiple phases

our surface we can utilise the time variation in order to get better beam forming control.

REFERENCES

- [1] Ma, Q., Xiao, Q., Hong, Q.R., Gao, X., Galdi, V. and Cui, T.J., 2022. Digital coding metasurfaces: From theory to applications. *IEEE Antennas and Propagation Magazine*, 64(4), pp.96-109.
- [2] Liu, S., Cui, T.J., Zhang, L., Xu, Q., Wang, Q., Wan, X., Gu, J.Q., Tang, W.X., Qing Qi, M., Han, J.G. and Zhang, W.L., 2016. Convolution operations on coding metasurface to reach flexible and continuous controls of terahertz beams. *Advanced science*, 3(10), p.1600156.
- [3] Zhang, L., Chen, X.Q., Liu, S., Zhang, Q., Zhao, J., Dai, J.Y., Bai, G.D., Wan, X., Cheng, Q., Castaldi, G. and Galdi, V., 2018. Space-time-coding digital metasurfaces. *Nature communications*, 9(1), p.4334.